

Infrared-to-visible albedo ratio of Phobos: Comparison with primitive asteroids

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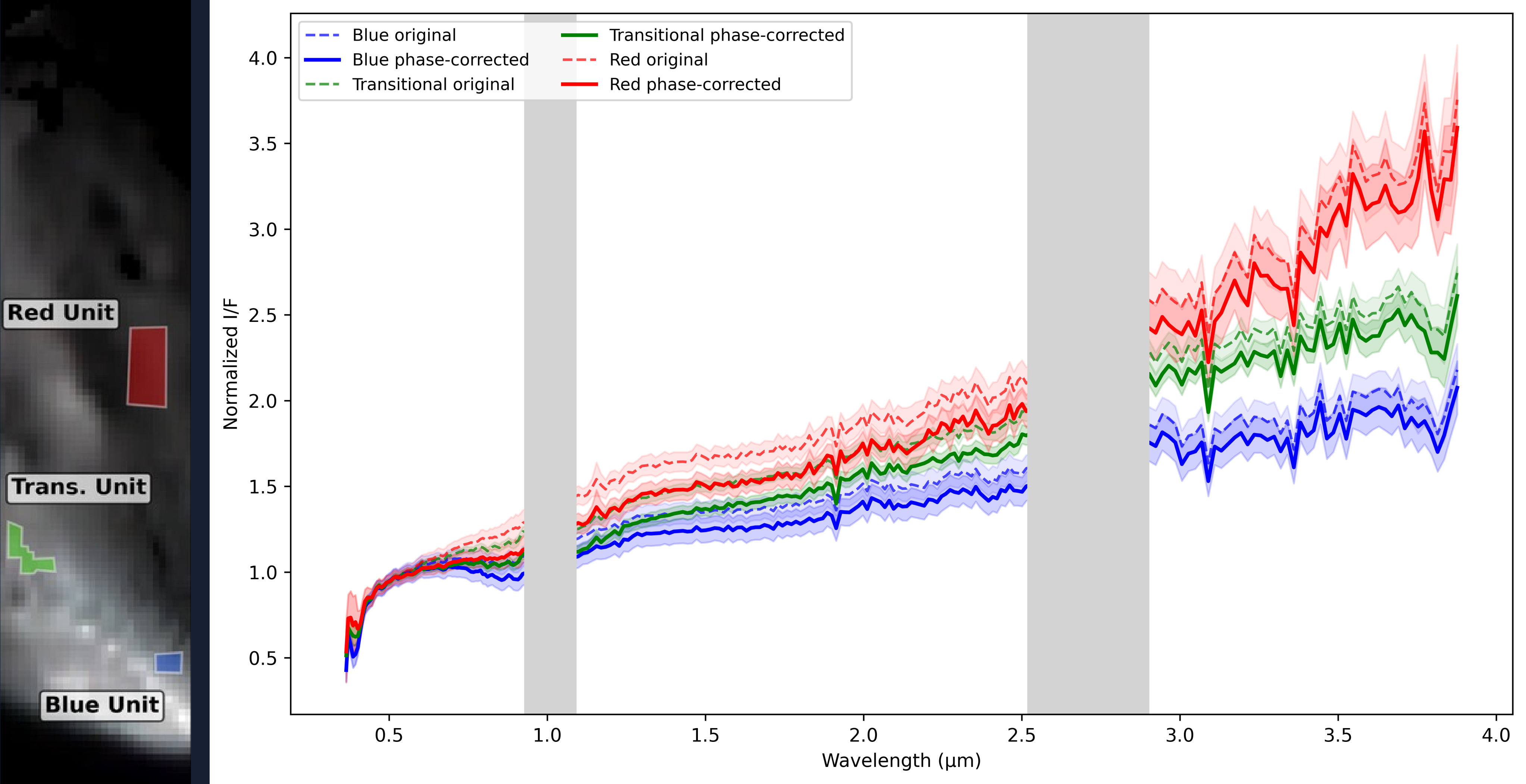
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Central question

Do the **Blue**, **Transitional**, and **Red** spectral units of Phobos occupy the same infrared-to-visible parameter space as primitive asteroid populations?

Phobos remains enigmatic: its surface is red-sloped and heterogeneous, yet its origin remains debated between primitive-body capture, impact-generated debris, and mixed-material scenarios [1,2].

OMEGA regions of interest and spectra



The selected Rols sample the Blue, Transitional, and Red units [1,3]. The spectra show a systematic increase in infrared-to-visible reflectance from Blue to Red. Because OMEGA observed Phobos at a high phase angle, the spectra were phase-corrected using CaSSIS-based phase-reddening estimates before comparison with asteroid populations [3,4].

Workflow

- OMEGA spectra**
Extract spectra for independently selected Blue, Transitional, and Red regions.
- Phase-reddening correction**
OMEGA observed Phobos at a high phase angle of 64°. A CaSSIS-based correction was applied to remove viewing-geometry reddening before comparing with asteroid populations.
- Asteroid catalogue**
Use MP3C visible and infrared albedos to compute asteroid $R_A = p_{IR}/p_V$ [5–7].
- Population comparison**
Compare Phobos R_P values against primitive asteroid R_A distributions.

Low-albedo filter	Band definitions
$p_V < 0.12$	$V = 0.50\text{--}0.60\text{ }\mu\text{m}$
	$W1 = 2.8\text{--}3.8\text{ }\mu\text{m}$
$R_P = \frac{\langle I/F \rangle_{W1}}{\langle I/F \rangle_V}$	$R_A = \frac{p_{IR}}{p_V}$

Spectral-unit (R_P) summary

Unit	R_P	Meaning
Blue ($R_{P,B}$)	1.922 ± 0.030	Least red
Transitional ($R_{P,T}$)	2.431 ± 0.018	Intermediate
Red ($R_{P,R}$)	2.981 ± 0.044	Reddest

Interpretation

The infrared-to-visible ratios trace the integrated spectral redness of each surface unit. Higher R_P values indicate redder broad-band behaviour, but they are not uniquely diagnostic of composition.

The observed progression from Blue to Transitional to Red may reflect differences in surface maturity, space weathering, exogenous material, and/or intrinsic compositional heterogeneity.

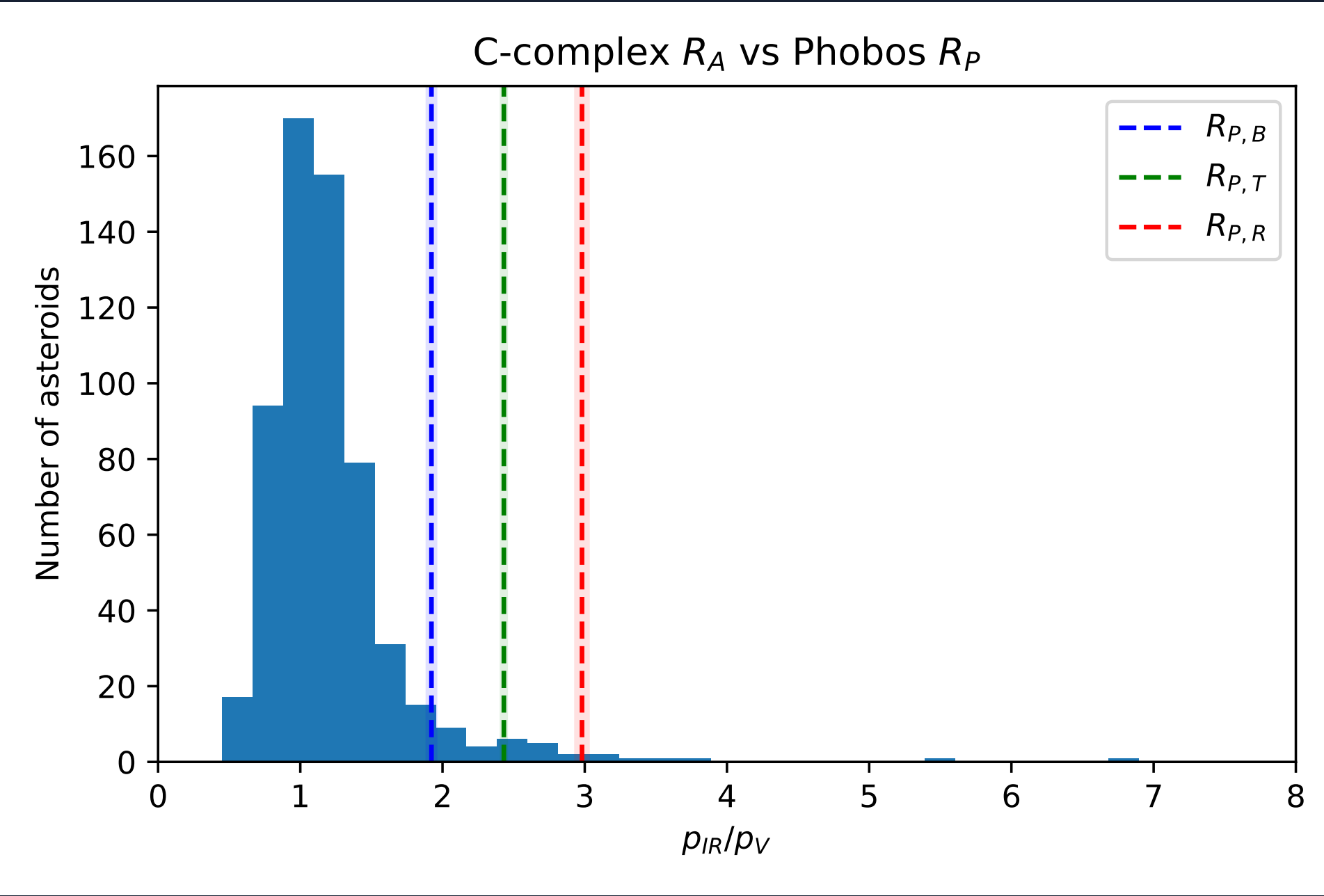
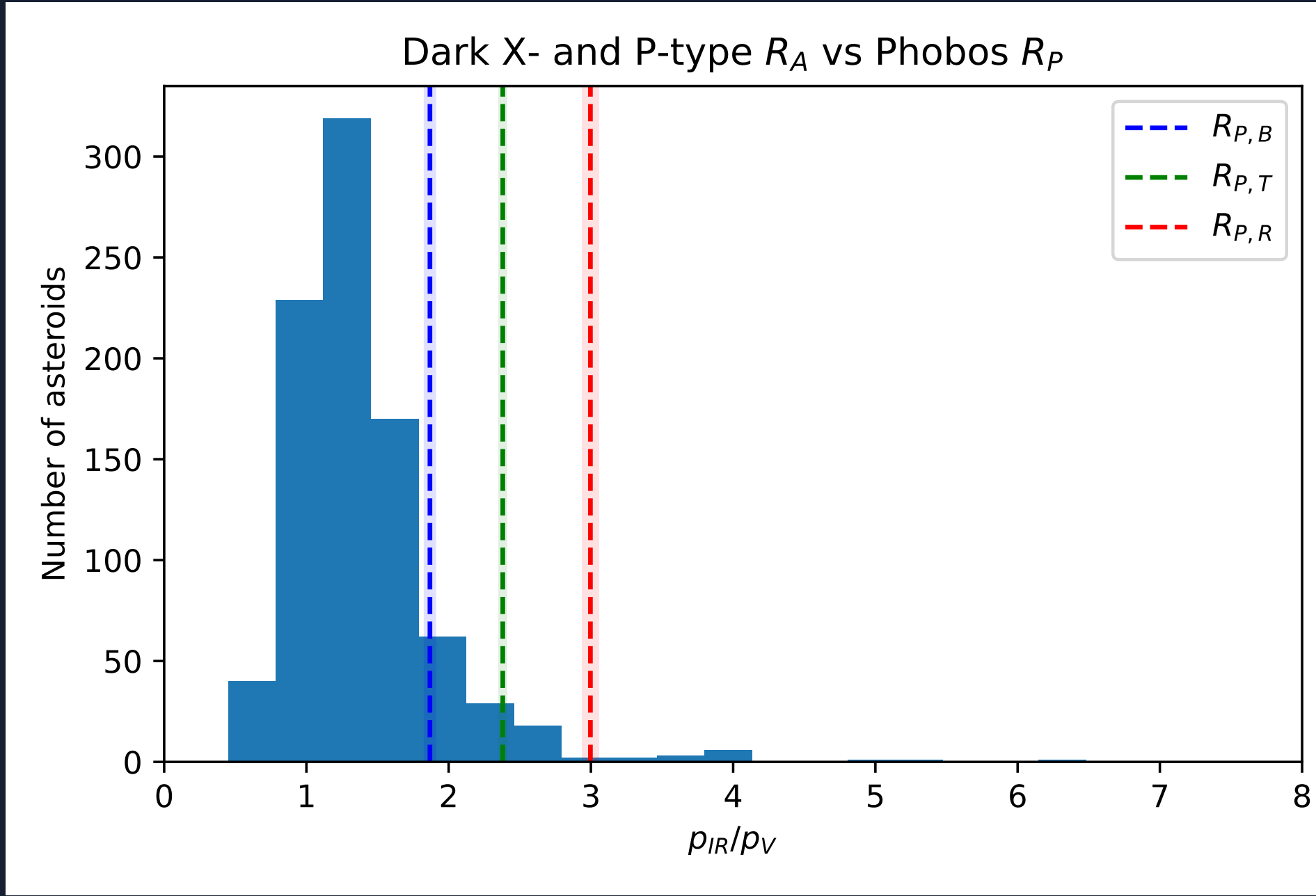
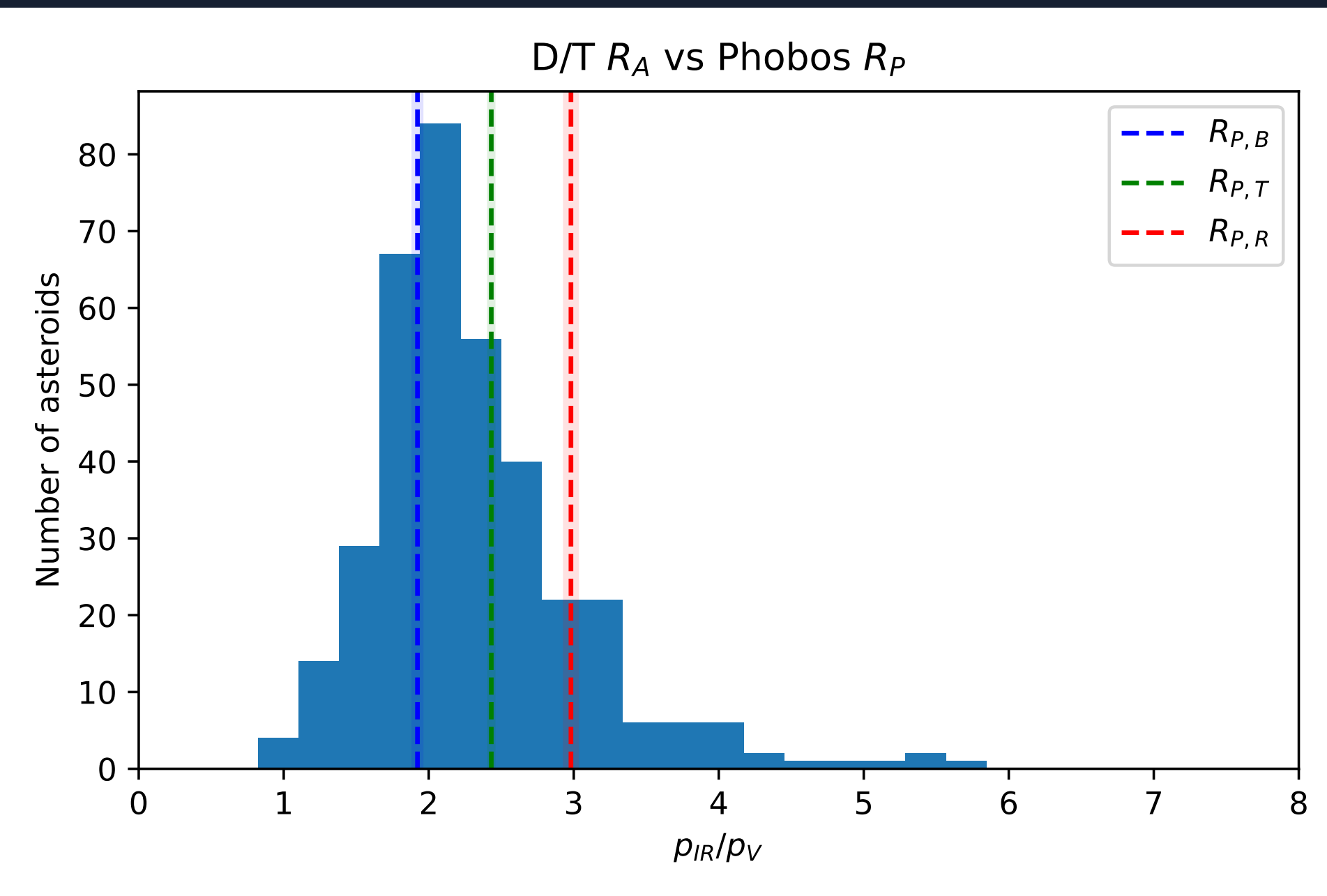
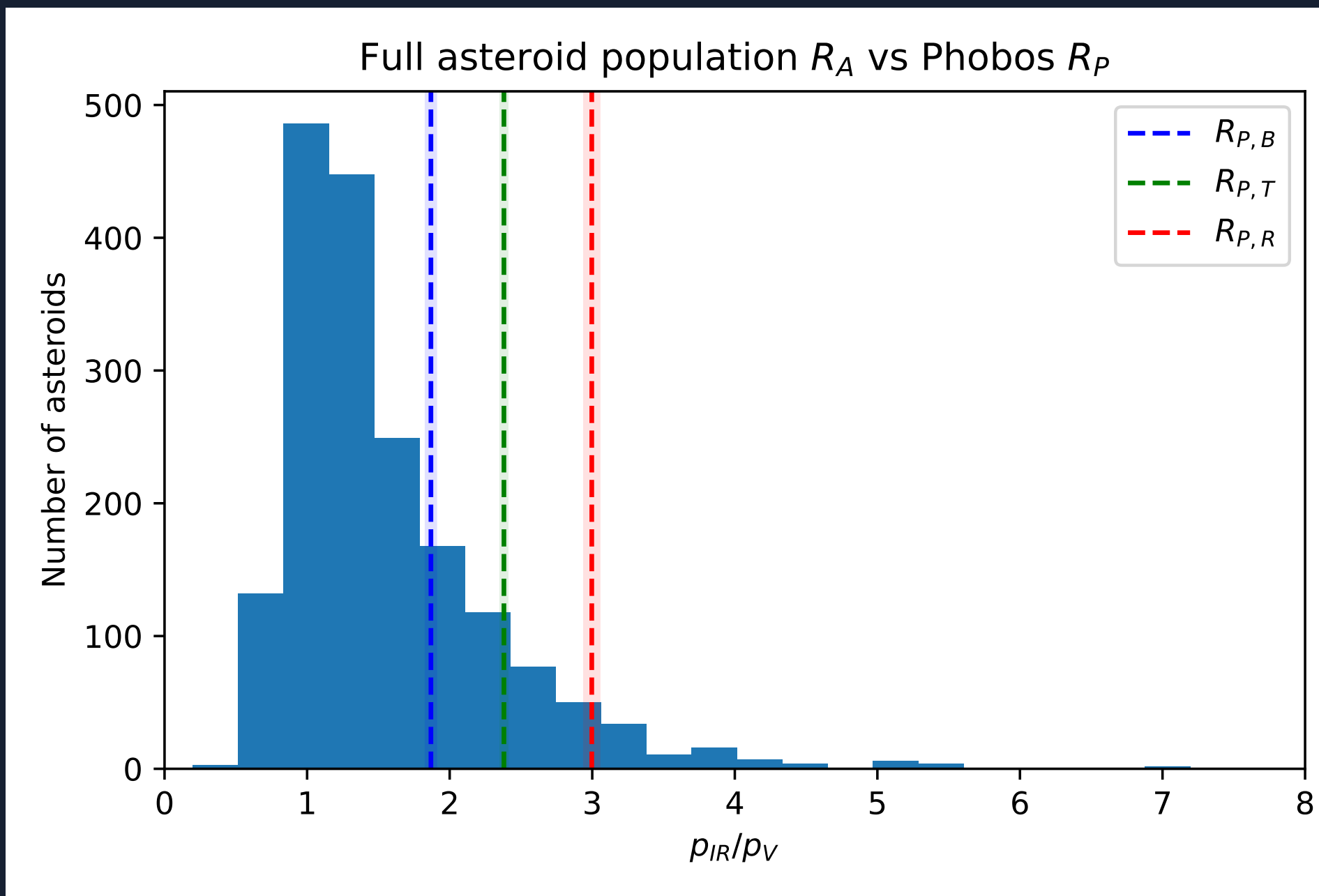
Phobos' spectral units span different parts of the primitive asteroid p_{IR}/p_V distribution.

- Blue, Transitional, and Red Phobos units all fall within the D/T-type asteroid p_{IR}/p_V range.
- The match is not uniquely compositional, but it supports Phobos as an empirical bridge between resolved spectroscopy and asteroid survey data.
- If Phobos was captured, its surface may be related to primitive D-, T-, or Z-type material [1,4,8].
- Redder units may indicate composition, surface maturity, or exposure of different parent-body depths.
- This is consistent with scenarios where P/T/D/Z-types represent outer primitive shells and some C-types sample more interior material [8].
- Spectral variation could also be due to implantation of exogenous material from darker carbonaceous asteroids [1,3].

Conclusions

- Infrared-to-visible ratios provide a compact comparison between Phobos spectral units and asteroid populations.
- R_P increases clearly from the Blue to Transitional to Red units.
- All three units overlap the D/T-type asteroid $R_A = p_{IR}/p_V$ range.
- Transitional and Red units extend into the high- R_A tail of primitive asteroid populations.
- This supports a possible link with red primitive material, while remaining degenerate with surface maturity.
- Future 3–5 μm MMX/MIRS observations will help test Phobos' origin and surface evolution [2].

Comparison to asteroid populations



The Phobos ratios overlap most strongly with red primitive asteroid populations. [4].

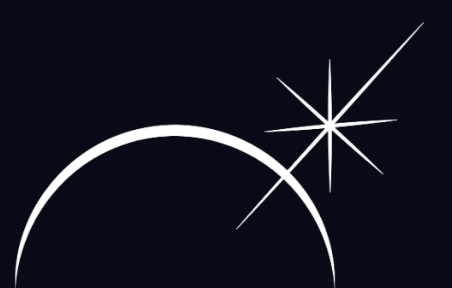
The Blue and Transitional units lie close to the D/T-type distribution, while the Red unit extends into the higher-ratio tail.

References

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